

Parag Agnihotri

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer and Data Scientist with 2+ years of progressive experience in designing and deploying scalable machine learning and Generative AI solutions. Demonstrated expertise in building production-ready RAG pipelines, NLP-driven applications, and Agentic AI systems while collaborating with cross-functional engineering teams to deliver business-focused AI products. Proficient in Python, Machine Learning, Deep Learning, LangChain, LangGraph, and modern LLM orchestration frameworks, with hands-on experience in model deployment, API development, and cloud-based AI workflows. AWS-certified professional focused on developing reliable, scalable AI applications that effectively bridge advanced research, data-driven insights, and real-world production deployment.

WORK EXPERIENCE

Associate Data Scientist

YASH Technologies Pvt. Ltd.

Aug 2024 – Present

On-site, Indore

- Designed and developed enterprise-scale AI/ML and Agentic AI solutions using LangChain, LangGraph, and CrewAI, building intelligent agents to automate complex business workflows.
- Built and deployed production-ready RAG pipelines and NLP applications, including Text2SQL systems, leveraging Milvus vector database for efficient semantic search and contextual retrieval.
- Developed custom LLM orchestration workflows, tool integrations, and context-aware execution pipelines with robust error handling, API integration, and observability mechanisms.
- Worked on enterprise AI products involving multi-agent collaboration, contextual reasoning, workflow automation, and intelligent task execution using modern Generative AI frameworks.
- Implemented prompt engineering strategies and retrieval optimization techniques to improve response relevance, contextual understanding, and overall LLM performance.
- Collaborated with cross-functional engineering and architecture teams to translate business requirements into scalable AI solutions while contributing to system design, deployment, and performance optimization.

Data Science Intern

Smartknower Edtech Solutions

Jan 2024 – Jun 2024

On-site, Indore

- Worked on industry-grade machine learning projects using Python libraries such as Pandas, NumPy, and Scikit-learn on large-scale educational datasets.
- Conducted exploratory data analysis (EDA) and statistical analysis to identify trends, patterns, anomalies, and improve data quality for downstream ML tasks.
- Designed end-to-end data preprocessing pipelines including data cleaning, feature engineering, feature selection, normalization, and transformation.
- Built, trained, and optimized machine learning models through hyperparameter tuning, cross-validation, and model selection techniques.
- Evaluated model performance using standard metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix to improve predictive accuracy.
- Collaborated with technical teams and stakeholders to translate business requirements into scalable, data-driven machine learning solutions.

PROJECTS

Unified Enterprise Solution - NIA

Data Scientist (Agent Development) & Multi-Agent System Designer

- Collaborated with a cross-functional team to build an intelligent multi-agent system for automating enterprise workflows.
- Designed and developed specialized AI agents using LangChain and LangGraph enabling modular orchestration of LLM-based workflows.
- Created custom MCP (Model Context Protocol) tools to enable agents to interact with external systems including APIs, databases, and file systems.
- Implemented Python-based data processing pipelines for handling structured and unstructured data within AI workflows.
- Built intelligent routing logic and task orchestration capabilities to enable coordination between multiple agents.

RapidRead – PDF to Podcast

Backend Developer & AI Pipeline Architect

- Built a Python-based AI pipeline that converts static PDF documents into conversational podcast-style audio using a multi-agent architecture powered by CrewAI.
- Implemented automated data processing workflows including PDF parsing, text extraction, semantic segmentation, and structured content transformation.
- Developed modular Python components for summarization, dialogue generation, and audio synthesis using Microsoft TTS.
- Built backend APIs using FastAPI and integrated a Streamlit interface for file upload, processing, and playback.

Gender Prediction using Acoustic Voice Data

Machine Learning Engineer

- Developed a machine learning system to predict gender from voice recordings using a dataset containing 10,000+ acoustic samples.
- Performed extensive exploratory data analysis (EDA) and feature engineering using pandas and NumPy to understand acoustic patterns.
- Implemented data preprocessing pipelines including cleaning, normalization, feature extraction, and train-test dataset splitting.
- Trained and compared multiple machine learning models including Decision Trees, Random Forests, and Support Vector Machines using scikit-learn.
- Evaluated model performance using metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix.
- Visualized data distributions and model results using Matplotlib to support model evaluation and analysis.

EDUCATION

Bachelor of Technology in CSE - Data Science

Acropolis Institute of Technology and Research, Indore

2020 – 2024

CGPA: 7.80

Higher Secondary (Class 12)

Brilliant Academy, Indore

2020

Percentage: 70%

SKILLS

Languages	Python, SQL
Machine Learning	Scikit-learn, NumPy, Pandas, Matplotlib, Data Preprocessing, Feature Engineering, Model Evaluation
Generative AI	RAG Pipelines, Agentic AI, LLM Orchestration, NLP, Text2SQL, LangChain, LangGraph, CrewAI
Backend	FastAPI, REST APIs, Streamlit
Databases	MySQL, Mongo DB, Vector Databases (Milvus, Qdrant)
Cloud	AWS, Cloud Computing, Serverless Architecture
Tools	Git/GitHub, Postman, API Dog
Core Concepts	Machine Learning, Deep Learning, Data Pipelines, Semantic Search, Prompt Engineering, AI Workflow Automation

CERTIFICATIONS

AWS Certified Generative AI Developer - Professional	View Badge
Amazon Web Services (AWS)	
AWS Certified Developer Associate	View Badge
Amazon Web Services (AWS)	
AWS Certified AI Practitioner	View Badge
Amazon Web Services (AWS)	